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Risk-Aware Template-Switching Control under Partial Quantum Observability

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Abstract

With only partial observations, a discrete reference template can fit the available data yet still be a harmful choice. We ask whether a test based on observed data can filter such template switches without access to the target state. An audit of our submitted three-qubit simulations found a degenerate clock-conditioned representation and a veto driven solely by an uncertainty threshold. We retain those results as an audit, without attributing their effect to relational spectral structure. R1 establishes an energy-matched finite-clock construction. R3 compares a continuous estimate with a direct nearest-template estimate and tests score, uncertainty, and held-out residual switching rules. Both candidates are fitted on four clock labels and scored unchanged on the other four. We evaluate one corrected run and three independent replications, each with separate reference-bank, calibration, and test sets. Because any one-sided veto reduces the unconditional number of harmful switches, we additionally compare selection quality at equal expected switching rates. In a reviewer-prompted analysis of the same four streams, retaining half of the score-only proposals by held-out residual lowers the harmful fraction among accepted switches by 20.8–23.7 percentage points relative to uncertainty ranking. The combined gate has a smaller, operating-point-dependent benefit. A continuous estimator optimized using all measurements nevertheless achieves higher grid-mean fidelity than these switching rules. The contribution is a controlled assessment of selection information and measurement cost in finite-data template switching. The clock specifies a physically consistent observation model; neither its necessity for the decision task nor a relational-spectral performance advantage is established.

1 Introduction

Quantum error correction relies on diagnostic information that can be incomplete or noisy. Syndrome-based correction, recovery maps, autonomous protection, and error mitigation address different parts of this problem [1, 2, 3, 4, 5]. Finite-shot state estimation raises a related decision question: when should a policy replace a continuous estimate with a discrete reference template? The template may fit the available counts yet yield lower target fidelity. We test whether counts withheld from fitting can filter these harmful switches. This decision test does not measure the efficacy of a syndrome decoder.

The submitted work proposed to derive a switching signal from clock-conditioned graph spectra. The audit reported here shows why that interpretation fails: the conditional density is constant, and the admissibility condition never binds. We document the submitted construction and test a held-out residual gate using an energy-matched finite clock and finite-shot counts.

Graph and spectral representations remain useful descriptive tools for the submitted audit [6, 7, 8]. The R3 safety test instead asks whether a template fitted on four clock labels predicts the other four as well as a continuous estimate. This finite-data decision is complementary to characterization and uncertainty quantification [9, 10, 11]. It does not evaluate a syndrome decoder or channel-level mitigation method.

R3 evaluates this question in a controlled encoded logical-qubit testbed. Candidate A is a continuous Bloch-ball projected least-squares estimate and candidate B is the direct nearest

template under the same fitting-label loss. A policy may accept or veto a proposed switch from A to B. The held-out gate can only veto a proposal; it cannot create a template switch. The comparison therefore exposes both harmful-switch reduction and any loss of mean fidelity relative to the continuous candidate.

R1 supplies an energy-matched finite-clock construction with nonconstant conditional states. R3 tests the switching decision with physical code noise, direct-template and uncertainty-only controls, and a corrected held-out residual gate. Each of its four fixed streams has separate bank, calibration, and test targets; uncertainty intervals resample target pairs and measurement seeds. The residual gate reduces harmful switches relative to score-only selection in every evaluated cell, but the subset relation alone cannot establish useful selection. R4 therefore compares the same records at equal expected switching rates and adds globally optimized continuous-loss estimators. This distinguishes information about proposal quality from a reduction in intervention frequency.

The contribution is a controlled benchmark of a specific decision: whether to replace a fitting-stage continuous estimate by a fixed-bank template. The held-out residual contains selection information beyond random rejection and uncertainty in this testbed, while the full-measurement continuous control exposes the cost of reserving data for that decision. R1 supplies the physically consistent observation model needed for the adopted clock interpretation; it is not a separate performance contribution. The validation principle is ordinary data splitting, applied here to a specified quantum measurement model with explicit candidate, information, and switching-rate controls.

2 Background and Positioning

2.1 Clock-conditioned relational dynamics

Relational formulations of quantum dynamics describe change through correlations with an internal clock rather than an externally supplied time parameter. This idea appears in background-independence and relational-dynamics discussions [12, 13]. The Page–Wootters mechanism gives the operational ingredient used here: a constrained global state can be conditioned on a clock subsystem to obtain relative system states [14, 15, 16]. Related work has studied robustness, interactions, noncausal circuits, measurement-related time flow, and computational uses of quantum time [17, 18, 19, 20, 21]. Our use of this literature is narrow. R1 uses clock-conditioned slices to construct a finite observation model, and R3 uses its labels to split fitting and validation records. The switching rules operate on the resulting count record. An equivalent randomized-setting experiment has the same likelihood, as shown in Section 6.1; the internal-clock interpretation is not required by the statistical decision rule.

2.2 Spectral graph representations of quantum structure

Graph-based and spectral representations provide compact descriptions of relational structure. In quantum information, graph Laplacians have been related to density matrices, separability, and quantum-state representations [6, 7, 22]. More broadly, quantum network models provide a language for connecting graph structure, correlations, and quantum dynamics [8]. These works motivate the use of spectral signatures as auditable summaries of relational connectivity.

The submitted work differed from graph-state or network modeling approaches in its use of clock-conditioned spectral families. A single graph spectrum was not the central object; each internal clock label produced a relational graph, and the ordered family of Laplacian spectra defined a relational spectral trajectory. The revision retains that construction only to make the failed submitted-grid mechanism auditable. It is not an evidence source for R3.

2.3 Quantum recovery, mitigation, and control

Quantum recovery and error mitigation have developed along several complementary directions. Recovery-map approaches formalize channel-level reconstruction and approximate reversibility [1], while quantum error-correction experiments and autonomous protection schemes demonstrate how encoded or stabilized systems can preserve quantum information under noise [23, 2, 24]. In the NISQ setting, mitigation methods such as probabilistic error cancellation, virtual distillation, detection-based mitigation, and learning-based correction address noisy circuit outputs without requiring full fault tolerance [3, 25, 26, 4, 5, 27].

The submitted framework placed a policy layer above two candidate recovery mechanisms. In those encoded-QEC simulations, syndrome-first recovery was the decoder-facing baseline, while relational admissible recovery supplied an alternative candidate. R3 asks a different, narrower question: it compares a continuous count-record estimate with a direct template under equal information and uses held-out counts to test the proposed switch.

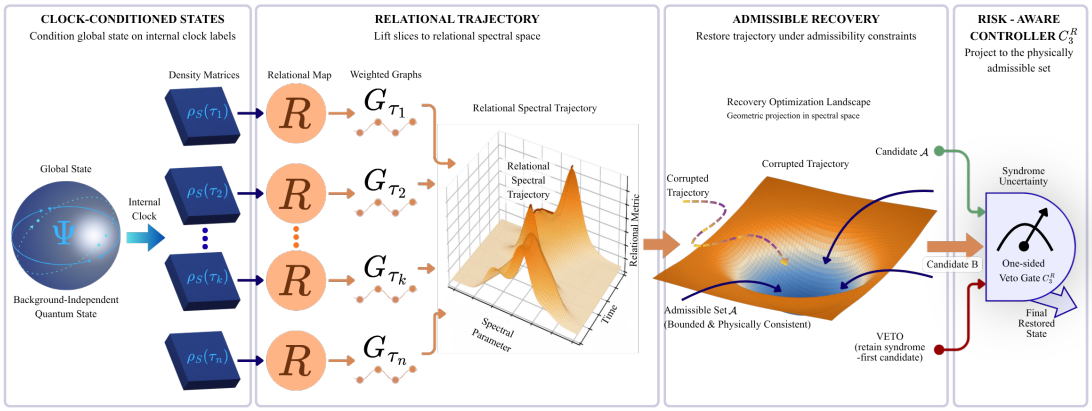


Figure 1. Submitted-pipeline overview, retained for audit reproducibility. A constrained global state is conditioned on internal clock labels to obtain system slices $\rho_S(\tau_r)$. Each slice is mapped to a weighted relational graph and reduced to a Laplacian spectral signature, yielding the relational spectral trajectory Γ_Ψ. The observed trajectory is restored through a reference-anchored admissible recovery objective, producing the relational candidate B. At the policy layer, C_3^R acts as a one-sided veto that can retain the syndrome-first candidate A or accept the C_2-proposed switch to B. The audit finds that this instantiated pipeline does not identify a relational-spectral efficacy effect.

2.4 Partial observability, uncertainty, and benchmarking

Reliable recovery decisions depend on how much diagnostic information is available and how trustworthy that information is. Benchmarking and characterization methods provide tools for quantifying device behavior and validating performance claims [9]. Randomized-measurement techniques and uncertainty-quantification frameworks also emphasize that finite, noisy, or incomplete observations must be interpreted with calibrated uncertainty [10, 11]. Noise modeling, live noise fingerprinting, robust control, and non-Markovian noise studies further show that quantum behavior can depend strongly on the structure and timing of noise processes [28, 29, 30, 31, 32, 33].

The submitted grid tested candidate selection when syndrome records were degraded, using a simple syndrome-stress index. Those diagnostics are retained for the audit. R3 instead tests template switching with finite-shot counts and a held-out residual. Its uncertainty statistic is a thresholded proxy, not a calibrated posterior. Both analyses report harmful switches and beneficial switches that the policy blocks.

3 Submitted framework and metrics (archival)

This section documents the framework used to generate the submitted grid so that the audit is reproducible. It is not the primary method supporting the revised claim. The audit in Section 4 establishes that the instantiated clock-conditioned trajectory is degenerate and that the submitted admissibility gate is nonbinding. Accordingly, the relational-spectral construction, recovery objective, and syndrome-stress policy below are historical implementation specifications rather than validated ingredients of R3 or R4. The corrected count-record method and the assessment of selection quality at equal switching rates are given in Section 6. Figure 1 therefore summarizes the submitted pipeline, not a validated recovery procedure.

3.1 Clock-conditioned relational formulation (submitted configuration)

The following formalism states the submitted intended construction. The audit shows that its actual implementation, specified below with $H_S = 0$, does not satisfy the required nondegenerate conditional-trajectory premise; it is retained only to define the historical artifact. We use a clock-conditioned relational formulation inspired by background-independent quantum dynamics and relational-time constructions [34, 14, 16]. The total Hilbert space is decomposed as

$$\mathcal{H}_{\text{tot}} = \mathcal{H}_C \otimes \mathcal{H}_S, \tag{1}$$

where $\mathcal{H}_C$ is an internal clock register and $\mathcal{H}_S$ is the system register. Rather than assuming an externally supplied time coordinate, we work with a constrained global state

$$\hat{H}_{\text{tot}}|\Psi\rangle = 0. \tag{2}$$

We use this global constraint not as a claim that the finite simulation realizes a complete Wheeler–DeWitt theory or quantum-cosmological dynamics [35, 36, 37], but as a construction

principle for generating internally indexed conditional system slices. In the computational pipeline, the role of the constraint is to define the admissible clock-conditioned family $\{\rho_S(\tau_r)\}_{r=1}^{m}$, from which all relational spectra, recovery objectives, and policy diagnostics are derived. In a finite-dimensional simulation, the existence and resolution of clock labels depend on spectral matching and recurrence constraints; those conditions are treated as consistency requirements and are moved to the Supplementary Information.

For a clock label τ, the clock-conditioned system state is

$$\rho_S(\tau) = \frac{\mathrm{Tr}_C[(|\tau\rangle\langle\tau|_C \otimes I_S)\,|\Psi\rangle\langle\Psi|]}{\mathrm{Tr}[(|\tau\rangle\langle\tau|_C \otimes I_S)\,|\Psi\rangle\langle\Psi|]}. \tag{3}$$

We therefore do not model recovery as a trajectory over an externally supplied background time coordinate. Instead, we condition a timeless global state on internal clock labels and construct recovery evidence from the resulting relational slices. Detailed consistency conditions for the global constraint, finite-dimensional clock representation, and clock-resolution assumptions are provided in Supplementary Note S1. In the implementation, each encoded simulation instance uses one clock qubit with initial state $|+\rangle_C$ and clock Hamiltonian $H_C = 0.8X_C$. The clock grid is uniform with

$$m = 9, \qquad \tau_r \in \{-2.0, -1.5, \ldots, 1.5, 2.0\}. \tag{4}$$

The conditional states $\rho_S(\tau_r)$ are materialized from the simulated global density matrix using Eq. (3). Unless otherwise stated, the same clock grid, clock Hamiltonian, and conditional-slice construction are used for all policies and stress protocols. This finite sampling is also why the framework is naturally compatible with partial observability: a degraded experiment may hide or corrupt part of the evidence available to the policy layer without changing the underlying definition of a clock-conditioned slice. The one-qubit clock is used deliberately as a minimal finite relational clock, not as a continuum-clock approximation [38, 17]. The intended question was whether clock-conditioned relational spectra could provide auditable recovery evidence. The audit shows that the submitted clock does not instantiate the nondegenerate conditional family needed to test that question; increasing clock resolution alone is not an established remedy.

3.2 Relational spectral representation (submitted configuration)

For each clock-conditioned state $\rho_S(\tau)$, we build a weighted relational network whose edge weights encode pairwise relational evidence between system degrees of freedom,

$$\begin{aligned} W_{ij}^{(\tau)} &= f(\rho_S(\tau); i, j), \\ W_{ij}^{(\tau)} &= W_{ji}^{(\tau)}, \qquad W_{ij}^{(\tau)} \geq 0. \end{aligned} \tag{5}$$

The submitted-grid experiments used explicit pairwise functionals rather than leaving f unspecified. Let $\rho_{ij}^{(\tau)}$ be the two-qubit reduction of $\rho_S(\tau)$ and let $\rho_i^{(\tau)}$ be the one-qubit reduction. For phase-flip-code experiments we use the absolute two-qubit coherence score

$$W_{ij}^{(\tau)} = \sum_{a \neq b} \left| (\rho_{ij}^{(\tau)})_{ab} \right|, \qquad i \neq j, \tag{6}$$

in the computational basis. For bit-flip-code experiments we use the von Neumann mutual-information score

$$W_{ij}^{(\tau)} = S(\rho_i^{(\tau)}) + S(\rho_j^{(\tau)}) - S(\rho_{ij}^{(\tau)}), \tag{7}$$

with negative numerical round-off clipped to zero. Alternative pairwise functionals are possible and are treated as representation-robustness questions for future work, but the submitted-grid policy comparisons use the fixed code-dependent mapping above. The two choices reflect the dominant relational signature of the corresponding encoded basis. Phase-flip protection is sensitive to phase coherence, so the computational-basis coherence functional directly measures off-diagonal relational structure. Bit-flip protection is associated with population-basis error structure, for which mutual information provides a basis-independent pairwise dependence score. Because this choice can affect sensitivity, we treat representation robustness as a limitation rather than claiming uniqueness of the graph functional. We use a graph Laplacian rather than raw edge weights because its spectrum summarizes collective connectivity, is stable under node relabeling, and allows perturbations to be compared through standard spectral quantities [39, 6, 7].

The corresponding degree matrix and graph Laplacian are

$$D_{ii}^{(\tau)} = \sum_{j \neq i} W_{ij}^{(\tau)}, \qquad L(\tau) = D^{(\tau)} - W^{(\tau)}. \tag{8}$$

Let the ordered Laplacian spectrum be

$$\begin{aligned} &\Lambda(\tau) = (\lambda_1(\tau), \ldots, \lambda_n(\tau)), \\ &\lambda_1(\tau) \leq \cdots \leq \lambda_n(\tau). \end{aligned} \tag{9}$$

The central object is not an individual graph or a single spectrum, but the clock-conditioned spectral family

$$\Gamma_\Psi : T \to \mathbb{R}^n, \qquad \Gamma_\Psi(\tau) = \Lambda(\tau). \tag{10}$$

We use the term trajectory to denote an ordered family over internal clock labels, not evolution with respect to external background time. This distinction matters for the recovery task. A local perturbation at one clock slice may be harmless if it is compatible with the neighboring slices, but the same perturbation can be suspicious if it creates a discontinuity in the spectral family. The object being recovered is therefore the whole clock-indexed spectral signature, not a collection of independent spectra. This representation is intended to expose relational inconsistencies that may not be captured by a single syndrome snapshot.

Computationally, the collection of admissible spectral families is treated as a finite metric trajectory representation. For two spectral families $\Gamma^{(a)}$ and $\Gamma^{(b)}$ sampled at clock labels $\{\tau_r\}_{r=1}^m$, we use

$$\begin{aligned} d_{\mathcal{M}}^{ab} &\equiv d_{\mathcal{M}}\left(\Gamma^{(a)}, \Gamma^{(b)}\right), \\ d_{\mathcal{M}}^{ab} &= \left(\sum_{r=1}^{m} w_r \left\|\Lambda^{(a)}(\tau_r) - \Lambda^{(b)}(\tau_r)\right\|_2^2\right)^{1/2}. \end{aligned} \tag{11}$$

The geometric terminology used below should be read as a formal interpretation of this finite metric trajectory representation. The detailed tangent-space, geodesic, and perturbation arguments are given in Supplementary Notes S2 and S3. The weights w_r allow the implementation to emphasize or de-emphasize particular clock slices, but the experiments reported here use the same metric form across all policies. Thus differences between C_1, C_2, C_3, and C_3^{R} arise from the decision layer, not from changing the underlying trajectory distance.

3.3 Admissible spectral recovery (submitted configuration)

Let Γ_{obs} be the observed spectral trajectory after noise or partial observation. Recovery is posed as a constrained restoration problem in spectral-trajectory space. We define an admissible set

$$\mathcal{A} = \mathcal{A}_{\mathrm{lap}} \cap \mathcal{A}_{\mathrm{rel}} \cap \mathcal{A}_{\mathrm{ref}}, \tag{12}$$

where $\mathcal{A}_{\mathrm{lap}}$ enforces Laplacian spectral constraints, $\mathcal{A}_{\mathrm{rel}}$ enforces relational smoothness and clock consistency, and $\mathcal{A}_{\mathrm{ref}}$ constrains compatibility with a reference family. In implementation these constraints are audited through penalty functions $\{\Phi_k\}_{k\in\mathcal{K}}$ with calibrated thresholds $\{\theta_k\}$. These three constraint classes play different roles. Laplacian constraints prevent spectral objects that cannot arise from an admissible relational graph. Relational smoothness penalizes abrupt clock-slice variation that is not supported by the reference dynamics. Reference compatibility prevents the recovery from drifting toward a structurally valid but physically irrelevant spectral family. We keep these as separate penalties rather than collapsing them into a single score so that later policy decisions can identify whether a candidate failed because of structure, relational continuity, or reference mismatch.

In the experiments, Φ_{lap} is the sum over clock slices of structural Laplacian violations: symmetry error, positive-semidefinite violation, row-sum violation, positive off-diagonal violation, and diagonal-consistency error. Φ_{smooth} penalizes slice-to-slice spectral jumps by summing $\|\Lambda(\tau_{r+1}) - \Lambda(\tau_r)\|_2^2/(\tau_{r+1} - \tau_r)^2$. Φ_{clock} compares spectral response between neighboring slices to the distinguishability of the corresponding clock states, and Φ_{ref} is the squared distance to the nearest reference trajectory. In the submitted implementation, thresholds were calibrated as $\mu + \kappa\sigma$ over the reference bank, with $\kappa = 1.5$. The associated calibration sweep is retained in the SI as historical implementation detail; it does not establish a discriminative admissibility gate on the submitted grid.

For a candidate trajectory Γ_X, the normalized admissibility violation is

$$v_X := \max_{k\in\mathcal{K}} \max\left\{0, \frac{\Phi_k(\Gamma_X) - \theta_k}{\theta_k + \epsilon}\right\}. \tag{13}$$

Thus $v_X = 0$ means that all active constraints are within threshold, while positive values measure the worst normalized excursion beyond the admissible boundary.

Admissibility defines which spectral families are physically and structurally acceptable. It does not by itself provide a restoration direction. We therefore introduce a reference-anchor term that explicitly pulls the recovered trajectory toward the nearest compatible reference family. Given a reference bank $\mathcal{R}$, the anchor is

$$\Gamma_{\mathrm{ref,anchor}} = \arg\min_{\Gamma' \in \mathcal{R}} d_{\mathcal{M}}(\Gamma', \Gamma_{\mathrm{obs}})^2 . \tag{14}$$

This anchor is the key distinction between feasibility and recovery. A projection onto $\mathcal{A}$ alone could return a trajectory that is merely allowed; it need not be the trajectory most compatible with the unperturbed relational behavior. The reference bank supplies a local restoration direction without requiring the method to know the clean trajectory of the test instance. In the encoded experiments, the bank contains three clean reference trajectories per code configuration: the corresponding logical reference state for the code configuration and two phase-rotated logical references at $\pm 20°$. The bank is generated with the same code, clock grid, clock Hamiltonian, and graph functional as the test row, but with the noise schedule removed. For each observed trajectory, the anchor is selected using only $d_{\mathcal{M}}(\Gamma', \Gamma_{\mathrm{obs}})$; no clean logical outcome, candidate fidelity, or row-level oracle information is used during anchor selection. The three-anchor bank is intentionally minimal. It tests whether a small auditable reference family can support recovery-side control, not whether a dense clean-state manifold has been approximated.

The recovery objective is

$$\begin{aligned} J_{\mathrm{rec}}(\Gamma) = {} & \alpha d_{\mathcal{M}}(\Gamma, \Gamma_{\mathrm{obs}})^2 \\ & + \beta d_{\mathcal{M}}(\Gamma, \Gamma_{\mathrm{ref,anchor}})^2 \\ & + \lambda_1 \Phi_{\mathrm{lap}}(\Gamma) + \lambda_2 \Phi_{\mathrm{smooth}}(\Gamma) \\ & + \lambda_3 \Phi_{\mathrm{clock}}(\Gamma). \end{aligned} \tag{15}$$

The first term keeps the recovered trajectory close to the observation, so the method does not discard the measured evidence. The second term pulls the solution toward the nearest compatible reference family, providing the actual restoration direction. The penalty terms then audit whether the resulting trajectory remains structurally and relationally admissible. This separation is deliberate: a candidate can be close to the observation but inadmissible, admissible but pointed toward the wrong reference family, or reference-compatible but too discontinuous across clock labels.

In the submitted-grid experiments, we use the discrete admissible projection as the default recovery operator because it is stable, interpretable, and directly auditable. Continuous convex refinement is treated as a diagnostic extension and reported in the Supplementary Information. Further details on admissibility penalties, reference-bank construction, anchor-weight calibration, and the two-stage recovery algorithm are provided in Supplementary Note S4.

3.4 Encoded-QEC protocol (submitted grid only)

Revision status: the configuration described in this subsection is the submitted grid, retained to make the audit reproducible. It has a degenerate conditional clock trajectory and cannot support an efficacy claim; its reported quantities are consequently labelled historical. The nondegenerate finite-clock check and the held-out switching experiment that support the revised scope are reported in Section 6 and in the supplementary section "Revision audit and new experiments."

The submitted encoded-QEC setting was designed as a controlled testbed for partial observability and unsafe recovery transitions. It does not support a decoder claim: the audit shows that its relational evidence is degenerate and its admissibility gate is inactive. The details below describe how complete, missing, corrupted, ambiguous, measurement-, and reset-error syndrome conditions were generated for the historical audit. R3 instead tests a two-candidate count-record decision problem with equal-information candidates.

Each submitted-grid simulation row produced two candidates:

$$A := \text{syndrome-first decoded recovery}, \qquad B := \text{relational admissible recovery}. \tag{16}$$

Candidate A was obtained by applying the code-specific syndrome recovery rule to the underlying noisy density. Candidate B was obtained by the admissible spectral recovery objective in Eq. (15). The policy then made the historical switch decision. As detailed in the SI, degraded syndrome symbols affected controller support rather than the candidate-A recovery channel. Both candidates are evaluated against the same clean logical trajectory when computing outcomes, but the policy

layer does not use that clean trajectory when making the decision. It sees only the candidate diagnostics, observed syndrome summary, admissibility reports, recovery objective, and uncertainty proxy. This separation is necessary because the intervention metrics later use the clean trajectory only as offline ground truth for analysis.

The submitted-grid experiments use three-qubit repetition codes in bit-flip and phase-flip configurations, with bit-flip, phase-flip, depolarizing, dephasing, and mixed-Pauli noise families. Each reported regime cell is averaged over seeds 11–20, giving ten random seeds per cell. The output-stem suffix `seed10` is a batch label for this ten-seed set. The encoding and syndrome-extraction circuits for both code configurations are shown in Supplementary Figs. S1 and S2. Syndrome observability is controlled by the observation ratio r_{obs}, syndrome-bit corruption probability p_{noise}, ambiguity level a_{amb}, measurement error probability p_{meas}, reset error probability p_{reset}, and measured syndrome information loss ℓ_{syn}. These quantities define the raw and clipped uncertainty proxies

$$u_{\mathrm{raw}} = (1 - r_{\mathrm{obs}}) + p_{\mathrm{noise}} + a_{\mathrm{amb}} + p_{\mathrm{meas}} + p_{\mathrm{reset}} + \ell_{\mathrm{syn}}, \qquad u_{\mathrm{syn}} = \mathrm{clip}_{[0,1]}(u_{\mathrm{raw}}). \tag{17}$$

The uncertainty proxy is a deliberately simple additive stress index rather than a calibrated posterior uncertainty. The clipped value is used by the decision gate so that $u_{\max} = 0.99$ implements a conservative saturation rule: saturated uncertainty blocks a proposed relational switch, while non-saturated uncertainty leaves the C_2 proposal eligible. The raw uncertainty is retained for diagnostics and bin-wise analysis. This distinction is important in the partial-plus-noisy and ambiguity-plus-measurement regimes, where several moderate degradations can combine into saturated decision uncertainty.

We compare four policies on the same candidates A and B. C_1 is a rule-based conservative hybrid that switches to B only when A is inadmissible and B is admissible, or when B is admissible, has lower recovery objective, and the observed syndrome is consistent. C_2 is the balanced proposal policy

$$S_X^{(2)} = \lambda_s s_X - \lambda_t d_X - \lambda_i (1 - a_X) + \lambda_o \frac{1}{1 + J_X}, \tag{18}$$

where a_X is the admissibility indicator, d_X is the trajectory distance to the observed trajectory, and s_X is the syndrome-support score. The current experiments use

$$(\lambda_s, \lambda_t, \lambda_i, \lambda_o) = (1.0, 1.0, 2.0, 0.5). \tag{19}$$

C_3 is a hard-admissibility ablation: if exactly one of A and B is admissible, it selects the admissible candidate; otherwise it falls back to a safety-weighted score. The initial C_3 policy was designed to test whether a globally strict safety-first region exists. Because this strict policy did not form an independent operating zone in the regime map, we use C_3^{R} as the revised risk-aware controller that preserves the safety motivation of C_3 while applying intervention only to proposed C_2 switches. This submitted policy sequence is intentionally cumulative. C_1 represents a conservative rule-based baseline. C_2 asks whether a balanced score can use relational evidence without excessive switching. C_3 tests whether a strict Boolean admissibility rule creates a distinct safety-first region. C_3^{R} is the submitted controller: it keeps C_2 as the proposal mechanism, but only allows its proposed switch to survive if additional safety and uncertainty gates pass.

The robust policy C_3^{R} treats C_2 as a proposal and applies a one-sided veto. Let $\Delta S = S_B^{(2)} - S_A^{(2)}$. The gate is

$$\begin{aligned} C_3^{\mathrm{R}} = B \iff\ & [C_2 = B] \\ & \wedge\, [\Delta S \geq \delta_S] \\ & \wedge\, [v_A \geq \delta_A] \\ & \wedge\, [B \text{ admissible}] \\ & \wedge\, [v_B \leq \delta_B] \\ & \wedge\, [u_{\mathrm{syn}} \leq u_{\max}]\,. \end{aligned} \tag{20}$$

Otherwise $C_3^{\mathrm{R}} = A$. The main operating point is

$$\begin{aligned} \delta_S &= 0, \qquad \delta_A = 0, \\ \delta_B &= 0, \qquad u_{\max} = 0.99. \end{aligned} \tag{21}$$

Thus C_3^{R} can only block C_2 switches; it never creates additional switches that C_2 rejected. The policy should therefore be interpreted as a downside-risk controller rather than a fourth independently tuned score rule. At the main operating point, the $v_A \geq \delta_A$ condition is nonrestrictive because $v_A \geq 0$ by definition. The active default gate therefore tests whether the C_2-proposed relational candidate is score-supported, exactly admissible within the calibrated tolerances, and not exposed to saturated syndrome uncertainty. We keep the v_A term in Eq. (20) because it is part of the general gate family and can be activated in ablations. At this operating point, the uncertainty gate is the only empirically active veto on the submitted grid. The exact-admissibility condition is intended to exclude candidates outside the calibrated admissible set, but no proposed candidate in this grid fails that condition. Complete policy maps, decision reasons, and operating-zone summaries are provided in Supplementary Notes S5 and S6. The clean-regime role of C_3^{R} is correspondingly limited. If syndrome evidence is reliable and the uncertainty gate is open, the robust policy should often reduce to C_2. Its value is tested primarily in regimes where syndrome evidence is partial, corrupted, or ambiguous enough that a proposed relational switch carries elevated downside risk.

3.5 Evaluation metrics and operating criteria

Since the controller is designed to suppress unsafe recovery transitions rather than maximize average fidelity alone, we evaluate both central performance and lower-tail risk. For each policy C, let

$$\Delta F_C = F_C - F_{\mathrm{before}} \tag{22}$$

be the logical-fidelity gain relative to the noisy observed trajectory before recovery. Central performance is summarized by the mean gain $\overline{\Delta F}_C$, the logical-success rate $P(F_C \geq F_{\mathrm{success}})$, and the non-worsening rate $P(\Delta F_C \geq 0)$. Unless otherwise stated, submitted-grid comparisons use

$$\varepsilon_F = 0.01, \qquad F_{\mathrm{success}} = 0.99. \tag{23}$$

These thresholds are fixed before policy comparison and are not tuned per regime. Mean gain alone is not sufficient for this study. A policy could improve the mean by taking aggressive relational switches while still creating rare but severe regressions. Because the controller is intended to regulate unsafe transitions, we require metrics that expose the lower tail of the gain distribution and the quality of the blocks performed by C_3^{R}.

Downside risk is summarized by the fidelity false-safe rate

$$\mathrm{FS}_F(C) = P[\Delta F_C < -\varepsilon_F]\,, \tag{24}$$

the empirical lower quantile $q_{0.05}(C)$, and the lower conditional value at risk

$$\mathrm{CVaR}_{0.05}(C) = \widehat{\mathbb{E}}\,[\Delta F_C \mid \Delta F_C \leq q_{0.05}(C)]\,. \tag{25}$$

A successful risk controller should remain inactive when syndrome evidence is reliable, but should reduce false-safe and lower-tail failures under partial observability. We also use oracle regret as an offline diagnostic. For each row, define

$$\begin{aligned} \Delta F^\star &= \max\{\Delta F_A, \Delta F_B\}, \\ \mathrm{Reg}(C) &= \Delta F^\star - \Delta F_C. \end{aligned} \tag{26}$$

This quantity does not define an implementable policy because it uses the row-level best candidate after the fact. It is nevertheless useful for measuring how far a policy falls from the best available candidate in the two-candidate decision problem.

For C_3^{R}, we additionally evaluate the quality of its interventions against the C_2 proposal. The block indicator is

$$B_R = \mathbf{1}[C_2 = B]\mathbf{1}[C_3^{\mathrm{R}} = A]. \tag{27}$$

A proposed C_2 switch is called harmful when

$$H_2 = \mathbf{1}[C_2 = B]\mathbf{1}[\Delta F_A > \Delta F_B + \varepsilon_F], \tag{28}$$

and beneficial when

$$G_2 = \mathbf{1}[C_2 = B]\mathbf{1}[\Delta F_B > \Delta F_A + \varepsilon_F]. \tag{29}$$

We report the block rate $P(B_R = 1 \mid C_2 = B)$, harmful-block precision $P(H_2 = 1 \mid B_R = 1)$, harmful-switch recall $P(B_R = 1 \mid H_2 = 1)$, and beneficial-switch block rate $P(B_R = 1 \mid G_2 = 1)$. Beneficial-switch retention is its complement. The utility balance is

$$E_{\mathrm{net}} = \sum_i B_R^{(i)} \max\{0, \Delta F_A^{(i)} - \Delta F_B^{(i)}\} - \sum_i B_R^{(i)} \max\{0, \Delta F_B^{(i)} - \Delta F_A^{(i)}\}. \tag{30}$$

Positive E_{net} means that, on the sampled grid, prevented harmful-switch losses outweighed missed beneficial-switch gains. It is a sample-level intervention diagnostic, not a theorem of dominance. We therefore also report normalized versions, including net effect per blocked switch and per C_2-proposed switch, in the Supplementary Information. Full metric definitions are given in Supplementary Note S7, and uncertainty-threshold diagnostics are given in Supplementary Note S8. The final operating criterion is therefore qualitative as well as quantitative: C_3^{R} should leave reliable regimes nearly unchanged, block a substantial fraction of harmful C_2 switches under degraded syndrome evidence, and maintain a positive net intervention balance after accounting for beneficial switches that it mistakenly blocks.

4 Submitted-grid audit

The submitted-grid analyses in this section are retained for transparency and for reproducibility of the originally reported tables and figures. They are not used as efficacy evidence in this revision. A code-and-data audit found that the submitted encoded construction has $H_S = 0$ while the clock is prepared in $|+\rangle_C$ with $H_C = 0.8X_C$. Its conditional density slices are consequently constant (up to numerical precision), and the nominal global state does not satisfy the stated zero-energy constraint. The same audit shows that, after removing the uncertainty conjunct ($u_{\max} = 1.00$), the score/admissibility-only counterfactual blocks no proposed switches on the submitted grid. Thus the tables below document the historical uncertainty-threshold effect; they do not demonstrate an independent relational-spectral admissibility effect. The differing historical row counts arose from different parameter-grid sizes, with ten seeds per combination: clean 240, partial 288, noisy-only 360, partial-plus-noisy 216, and ambiguity-plus-measurement 360 combinations. Supplementary Table S13 gives bootstrap intervals for the historical comparison, and Table S14 gives the signed effects of individual blocked switches using the same harm tolerance as the intervention metrics.

4.1 Operating-zone structure of the baseline policies

The submitted clean-syndrome regime map classified baseline policies into the following operating zones. The clean map contains 240 regime cells, each averaged over ten random seeds. As summarized in Table 1, C_1 is the preferred policy in most cells, while C_2 occupies a smaller but structured correction zone. The strict C_3 policy does not form an independent preferred region.

This baseline structure is descriptive only. Since B was admissible in every submitted row and the score/admissibility-only ablation has zero blocks, it does not identify a useful hard-admissibility veto. The submitted C_3^{R} effect is instead an uncertainty-conditioned veto applied to proposed C_2 switches. The C_2-preferred cells are localized to phase-flip-code rows, with 16 dephasing, 10 phase-flip, 9 mixed-Pauli, 6 depolarizing, and 2 bit-flip cells. We use the clean map only to identify whether the original policy family contains distinct baseline operating zones. Degraded-regime behavior is evaluated separately through the C_2 versus C_3^{R} comparison below.

Table 1. Baseline policy operating-zone summary under reliable syndrome evidence. Preferred-policy counts are computed over 240 clean regime cells, with ten random seeds per cell.

Policy	Cells	Share
C_1	197	0.8208
C_2	43	0.1792
C_3	0	0.0000
C_3^{R}	0	0.0000

4.2 Regime-level effect of C_3^{R}

The regime-level comparison between C_2 and C_3^{R} is shown in Table 2. The most important pattern is selectivity: C_3^{R} remains inactive in the clean and noisy-only controls, but intervenes in partial and partial-plus-noisy syndrome regimes.

In the clean map, C_3^{R} leaves the C_2 behavior unchanged. The controller blocks none of the 1529 rows where C_2 selected the relational candidate. In the noisy-only map, it also remains inactive: C_2 and C_3^{R} have identical false-safe rate, lower-tail metrics, and logical success rate. This provides a negative control. The robust controller does not treat syndrome-bit corruption without observation loss as sufficient reason to impose a universal conservative bias. The noisy-only regime is not risk-free; its false-safe rate remains comparable to the partial regimes. However, because the observation ratio is not reduced and the uncertainty index does not saturate, C_3^{R} does not activate. This clarifies that the default controller is targeted to saturated partial-observability stress rather than to all forms of recovery risk.

Table 2. Historical submitted-grid comparison between C_2 and C_3^{R}. Each entry is reported as C_2/C_3^{R}. The table documents the uncertainty-threshold effect; it is not evidence for an admissibility mechanism.

Regime	Mean	Success	Non-wors.	FS_F	$q_{0.05}$	$\mathrm{CVaR}_{0.05}$
Clean	0.1273/0.1273	0.6358/0.6358	0.9554/0.9554	0.0238/0.0238	0.0000/0.0000	-0.0033/-0.0033
Noisy-only	0.1169/0.1169	0.4556/0.4556	0.8814/0.8814	0.0678/0.0678	-0.0104/-0.0104	-0.0207/-0.0207
Partial	0.1169/0.1189	0.4552/0.5976	0.8774/0.9306	0.0694/0.0389	-0.0152/-0.0006	-0.0208/-0.0147
Partial+noisy	0.1170/0.1212	0.4551/0.7241	0.8792/0.9755	0.0685/0.0134	-0.0104/0.0000	-0.0206/-0.0034
Ambig.+meas.	0.1600/0.1604	0.5208/0.5978	1.0000/1.0000	0.0000/0.0000	0.0192/0.0238	0.0078/0.0084

4.3 Downside-risk reduction under partial syndrome observability

The historical uncertainty-threshold effect appears in the lower tail rather than in a large shift of the mean. In the partial-syndrome regime, the mean gain changes modestly from 0.1169 to 0.1189, while the false-safe fidelity rate decreases from 0.0694 under C_2 to 0.0389 under C_3^{R}. The lower 5% quantile moves from -0.0152 to -0.0006, while $\mathrm{CVaR}_{0.05}$ improves from -0.0208 to -0.0147. The logical-success rate increases from 0.4552 to 0.5976, and the non-worsening rate increases from 0.8774 to 0.9306.

The effect is stronger under partial-plus-noisy syndrome observability. There, C_3^{R} changes the mean gain from 0.1170 to 0.1212, but reduces the false-safe fidelity rate from 0.0685 to 0.0134 and shifts $q_{0.05}$ from -0.0104 to 0.0000. The lower-tail $\mathrm{CVaR}_{0.05}$ improves from -0.0206 to -0.0034. Thus the controller does not primarily act by maximizing average gain; it suppresses the negative tail of the recovery outcome distribution. This pattern is visualized in Fig. 2. The reduction should be interpreted together with the uncertainty-bin analysis in Supplementary Note S8: most blocks occur in saturated-stress rows, so the observed lower-tail improvement reflects a conservative veto of C_2 switches under high partial-observability stress. Because the score/admissibility-only counterfactual has zero blocks, these historical improvements cannot be assigned to the relational admissibility condition. They are retained to document the submitted uncertainty-threshold behavior only.

The ambiguity-plus-measurement regime is informative for a different reason. Both C_2 and C_3^{R} have zero false-safe fidelity rate in this regime, so the binary false-safe metric alone would miss any distinction. The lower-tail metrics and logical-success rate nevertheless improve slightly under C_3^{R}, indicating that false-safe rate should be interpreted together with lower-tail and intervention diagnostics.

4.4 Intervention quality: harmful blocks, missed gains, and net effect

The previous subsection records historical submitted-grid downside-risk values. Because C_3^{R} can only veto proposed C_2 switches to B, the relevant quantities are conditional on $C_2 = B$.

In the partial-syndrome regime, C_2 selected B in 1094 rows and C_3^{R} blocked 476 of those switches, a conditional block rate of 0.4351. Among the blocked switches, the harmful-block precision was 0.8697: most blocked switches were cases where staying with A would have produced larger fidelity gain than switching to B. The prevented loss was 12.2670, while the missed beneficial gain was 6.6055, yielding a positive net intervention effect of 5.6616.

In the partial-plus-noisy regime, the intervention is stronger. C_3^{R} blocked 670 of 821 proposed C_2 switches to B, with harmful-block precision 0.8791 and harmful-switch recall 0.8475. The prevented loss was 17.4350 and the missed gain was 8.3139, giving a net intervention effect of 9.1212. These values describe the observed intervention balance: prevented loss exceeds missed gain on this grid. They do not isolate selection quality from the number of vetoes; R4 supplies that comparison at equal expected switching rates. The conditional intervention rates, aggregate balance, and normalized net effect across the active regimes are summarized in Fig. 3.

The cost of the controller is also visible. C_3^{R} blocks some beneficial switches: the beneficial-switch block rate is 0.3750 in the partial regime, 0.6739 in the partial-plus-noisy regime, and 0.4211 in the ambiguity-plus-measurement regime. The observed benefit therefore should not be described as free improvement. It is a risk-control tradeoff in which missed gains are accepted only when the prevented losses are larger on the sampled grid. The high beneficial-switch block rate in the partial-plus-noisy regime shows that C_3^{R} is not an efficiency-maximizing selector. Its operating point is intentionally conservative and should be evaluated by net downside-risk reduction, including normalized net effect, rather than by switch acceptance alone. This cost is substantial in the partial-plus-noisy regime, where more than half of beneficial C_2 switches are also vetoed. The present operating point should therefore be viewed as a safety-biased baseline rather than a tuned optimal controller.

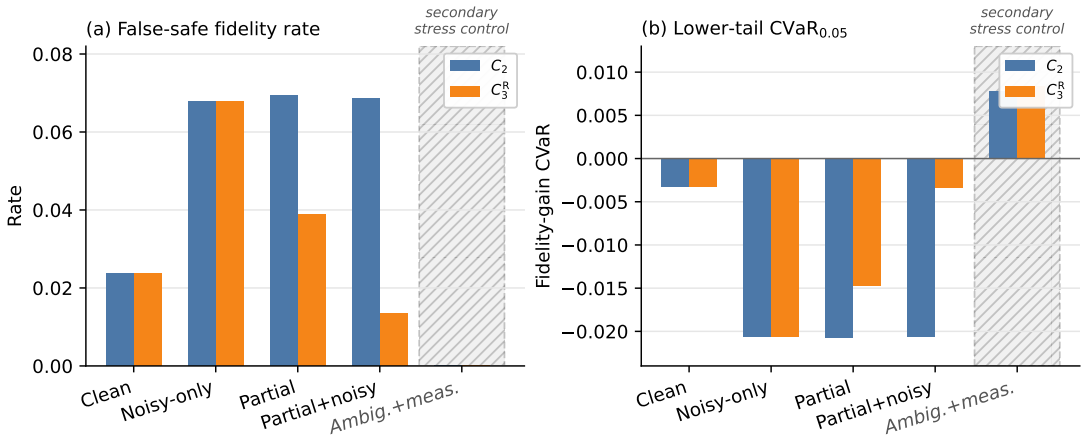


Figure 2. Historical submitted-grid downside-risk values for C_2 and C_3^{R}. Panel (a) shows the false-safe fidelity rate FS_F; panel (b) shows lower-tail fidelity gain $\mathrm{CVaR}_{0.05}$. The patterns are attributable to the submitted uncertainty gate and are not used as evidence for a relational-spectral mechanism.

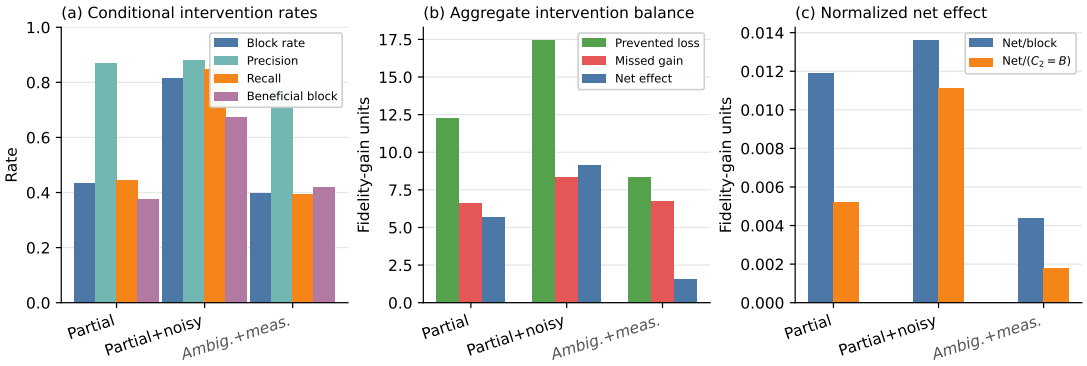


Figure 3. Historical submitted-grid intervention diagnostics for C_3^{R} in regimes where it actively blocks C_2 switches. Panel (a) reports the conditional intervention rates: block rate (given $C_2 = B$), harmful-block precision, harmful-switch recall, and beneficial-switch block rate. Panel (b) shows the aggregate intervention balance, comparing prevented harmful-switch loss, missed beneficial-switch gain, and the resulting net effect, all in fidelity-gain units. Panel (c) normalizes the net effect per blocked switch and per C_2-proposed switch to separate aggregate benefit from the number of vetoed proposals. The ambiguity-plus-measurement regime is hatched as a secondary stress control.

4.5 Negative controls and observed operating limits

The clean and noisy-only regimes serve as negative controls. In both cases, C_3^{R} is effectively inactive and does not materially change the C_2 outcomes. This is desirable: a risk-aware controller should not impose a universal conservative bias when the syndrome evidence is either reliable or only mildly corrupted without partial observability. The raw uncertainty mean is 0 in the clean map and 0.038 in the noisy-only map, far below the saturation condition that triggers uncertainty-based veto behavior.

The submitted partial-plus-noisy and partial rows have the largest uncertainty-threshold effects. The ambiguity-plus-measurement regime provides a weaker descriptive contrast: C_3^{R} changes logical-success rate from 0.5208 to 0.5978 and has positive net intervention value, while the binary false-safe rate is already zero for both policies. None of these submitted-grid patterns identifies a relational-spectral mechanism or a decoder advantage. Threshold diagnostics further show that setting $u_{\max} = 1.00$ removes the veto and recovers C_2 behavior, whereas stricter $u_{\max} = 0.75$ increases blocking and can reduce FS_F further at the cost of more beneficial-switch loss. In some regimes this becomes overly conservative. Thus the default point is not claimed to be optimal; it is a conservative operating point that exposes the downside-risk tradeoff.

5 Interpretation of the submitted-grid audit

5.1 What the submitted grid does and does not show

The submitted rows establish only a descriptive operating pattern: the uncertainty gate is inactive in clean and noisy-only rows and active in the partial-observability rows. They do not establish

that relational spectral evidence is useful, because the clock-conditioned densities are constant and candidate B is admissible in every row. The audit separates this negative finding from the new R3 held-out residual test below.

The strict-C_3 result is correspondingly non-informative: its condition does not create an independent preferred zone, and the removed-uncertainty ablation does not veto a single proposed switch. We withdraw the former interpretation of this observation as evidence for a more elaborate admissibility controller.

5.2 Partial observability and decision geometry

The submitted decision variables include a score difference, an admissibility flag, normalized constraint violations, and an additive syndrome-stress index. They are useful for reproducing the historical switch decisions, but their geometry cannot be interpreted as a nondegenerate relational trajectory because the conditional density is constant.

At the submitted default setting, the only empirically active conjunct is the uncertainty saturation guard. The policy is mathematically one-sided—it can remove candidate-B switches from the C_2 decision set but cannot create new switches—yet this fact alone does not supply a physical or spectral mechanism.

The historical intervention metrics quantify the cost of that uncertainty-only veto, including missed beneficial switches. In particular, an unconditional harmful-switch reduction is guaranteed by the subset relation and is not sufficient evidence of selective rejection. They are not used to claim a mechanism or to optimize a gate across codes and noise models.

5.3 Limits of the submitted recovery comparison

The submitted candidate comparison is not a damaged-syndrome decoder benchmark. Candidate A used the complete circuit syndrome-recovery channel applied to the underlying noisy density; damaged, missing, and ambiguous syndrome symbols affected only the controller support. Candidate B is a discrete template candidate, and its submitted admissibility flag never rejects a row.

The historical two-candidate values therefore do not answer a QEC recovery question. R3 replaces the information asymmetry with a count-record setting and explicitly compares the template candidate to a continuous equal-information estimate. It retains false-safe rate as a selection-risk metric but does not claim a decoder advantage.

The binary template-switching geometry remains a deliberately limited task. It makes conditional block and false-safe metrics computable, but a multi-candidate or channel-level comparison requires a separate design.

5.4 Clock-conditioned interpretation

The submitted Page–Wootters-type construction is not a valid finite relational trajectory for the stated initial clock and Hamiltonians. R1 supplies a valid energy-matched finite-clock data construction, while R3 uses its clock labels as split fitting and withheld validation records. Neither construction is offered as a foundations-level result about quantum time.

5.5 Limitations and outlook

The remaining limitations are substantive. Larger codes, broader channel families, optimized syndrome decoders, and multi-candidate recovery sets are needed before making decoder-level claims [40, 41, 42]. R3 is a controlled two-candidate selection test, not a competitive QEC benchmark.

Alternative structural summaries, larger template banks, multipartite witnesses, and learned representations may change the R3 switching sensitivity and need independent tests [43, 44, 45].

The current uncertainty variable is not a calibrated posterior. In the submitted grid it is the only observed active gate, while in R3 it is one of two fixed calibrated switching thresholds. Future work should calibrate uncertainty, compare structural summaries, and test the task beyond the sampled finite-data settings [46, 47].

Taken together, the submitted-grid audit does *not* support the previous mechanistic attribution to relational spectral admissibility. It motivates an explicit observation model and an assessment of the gate beyond its one-sided veto property, which are provided next.

6 Observation model and template-switching assessment

6.1 Physically specified finite-clock construction (R1)

The revision experiments use an encoded logical qubit and an energy-matched two-level clock, rather than the submitted $H_S = 0$ setup. Let

$$H_C = \omega Z_C, \qquad H_S = -\omega(n_x X_L + n_z Z_L), \qquad \omega = 0.8, \tag{31}$$

where the tilted setting is $n = (1, 0, 1)/\sqrt{2}$. If $|n_+\rangle$ and $|n_-\rangle$ are the logical eigenstates of $n_x X_L + n_z Z_L$ and $|\psi\rangle = c_+|n_+\rangle + c_-|n_-\rangle$, we prepare

$$|\Psi\rangle = c_+|0\rangle_C|n_+\rangle_L + c_-|1\rangle_C|n_-\rangle_L, \tag{32}$$

which obeys $(H_C + H_S)|\Psi\rangle = 0$. Eight clock effects $E_k = (2/8)|\tau_k\rangle\langle\tau_k|$ at $\tau_k = k\pi/(8\omega)$ form the finite POVM. R1 verifies the null constraint, conditional evolution, POVM completeness, nondegenerate clock geometry, and the rank-three tilted observation design. It is an observability qualification, not a QEC efficacy result.

The physical role of this construction must be separated from the statistical role of the gate. For the population-count readout used here, the likelihood of clock label k and outcome $z \in \{-1, +1\}$ can be written

$$p(k, z \mid \boldsymbol{r}) = \frac{1}{16}\left[1 + z(1 - 2\eta)\boldsymbol{d}_k^{\mathsf{T}}\boldsymbol{r}\right], \tag{33}$$

where $\boldsymbol{r}$ is the initial logical Bloch vector, η is the known readout-flip probability, and $\boldsymbol{d}_k$ includes the declared rotation and physical noise. Selecting one of eight ordinary measurement settings uniformly and measuring its corresponding logical population yields the same likelihood. Thus energy matching is required for our chosen stationary-state clock interpretation, but no clock-specific statistical advantage or operational necessity follows. Graph spectra do not enter R3 or R4.

6.2 Corrected held-out template-switching risk test (R3)

R3 uses four fixed data streams, each with disjoint reference-bank, calibration, and held-out target sets. Stream 0 reuses R2's random streams to isolate the gate-definition correction; streams 1–3 use independent bank, calibration, target, and measurement streams. The switching policy receives finite-shot count records, the known dynamics, and a fixed 32-state bank. Four clock labels are used for fitting, and the other four for validation. The bank, calibration, and test targets are pure logical-qubit states. Dynamics, physical noise, and readout-flip probabilities are known to the estimators. Candidate A is a continuous Bloch-ball projected weighted least-squares estimate; candidate B is the direct nearest template under the same fitting-label loss.

Let $L_{\mathrm{val}}(X)$ be the weighted squared loss of the unchanged fitting-label candidate X on the validation-label counts. The held-out residual statistic is

$$h = L_{\mathrm{val}}(B_{\mathrm{fit}}) - L_{\mathrm{val}}(A_{\mathrm{fit}}). \tag{34}$$

Neither candidate is refit on validation labels. Score-only C_2, uncertainty-only U, held-out-residual-only H, and the combined gate C_3^R are evaluated on identical records. For each stream, the score, uncertainty, and residual thresholds are set once to the 75th empirical quantiles of that stream's separate calibration set at the fixed bit-flip, 1,024-copy, physical-noise 0.05, readout-flip 0.05 operating point. They remain fixed across all 24 held-out test cells in that stream; their values may differ between streams. The target density is used only after a decision to calculate fidelity. An oracle nearest-template fidelity is reported only as an offline bank-resolution diagnostic, never as a policy input.

In every one of the 24 grid cells in each of the four streams, $H - C_2$ has a negative harmful-switch-rate difference whose two-way target-pair/measurement- seed 95% interval excludes zero. At the fixed middle operating point (bit-flip, 1,024 prepared copies, physical-noise probability 0.05, and readout flip 0.05), the $H - C_2$ differences are -14.844, -8.333, -10.677, and -5.990 percentage points in streams 0–3. The two-way interval for $C_3^R - U$ is negative in 12, 8, 12, and 6 cells, respectively; the combined gate therefore does not improve uniformly on uncertainty-only selection. The sign of the $C_3^R - A$ mean-fidelity difference also varies across streams. Full results and intervals are reported in the Supplementary Information.

These unconditional harmful-switch reductions are descriptive: H accepts a subset of the C_2 switches, and C_3^R a subset of the U switches. They cannot alone establish that the rejected proposals are selectively worse. R4 addresses this issue directly.

6.3 Selection quality at equal switching rates (R4)

R4 is a reviewer-prompted secondary analysis of the existing R3 streams. Its analysis rules were fixed before computing the new comparisons, after the original R3 outcomes were known. It replays all 73,728 target/count records from their recorded seeds and verifies agreement with the saved R3 diagnostics and fidelities; no new count realizations are introduced. Within each cell, we first thin the original C_2, H, and U decisions uniformly to the expected number accepted by C_3^R. Averaging over this randomization gives identical expected switching rates. Only the twelve 1,024-copy cells per stream have positive common coverage; at 256 copies, the original U and C_3^R accept no switches. The harmful fraction among accepted switches, rather than its unconditional count, now measures selection quality. Paired two-way bootstrap samples recompute the thinning factors.

At common coverage, the stream-mean harmful fraction for H is lower than for uniformly thinned C_2 by 7.62, 3.00, 6.19, and 4.48 percentage points in streams 0–3. The corresponding $C_3^R - U$ reductions are 5.23, 1.46, 4.37, and 1.93 points. All eight paired pointwise 95% intervals exclude zero (Supplementary Table S17). These contrasts establish selection information in the evaluated positive-coverage cells, beyond the automatic effect of accepting fewer switches.

We also compare batch rankings that accept 25%, 50%, or 75% of the original C_2 proposals. Scores are fitting loss difference, uncertainty u, residual h, and the combined statistic $\max(u/t_u, h/t_h)$ using the original R3 calibration thresholds. Uniform random retention supplies a control. Boundary ties are randomized analytically, so expected switching rates match exactly in each cell and resample. Cutoffs use the batch's observed diagnostics without target outcomes; these are batch selection curves, not newly calibrated deployment rules. Figure 4 shows that residual ranking is more selective than uncertainty or their conjunction over all three retention levels in the stream averages. At 50% retention, its $H - U$ harmful-fraction differences are -20.78, -21.76, -23.68, and -23.50 percentage points, with all four paired intervals below zero.

The numerical harm criterion $F_B < F_A - 10^{-12}$ counts even very small losses. A follow-up sensitivity analysis keeps every decision rule fixed and applies absolute fidelity margins 0.001, 0.005, and 0.01. At margin 0.01 and 50% retention, the $H - U$ harmful-fraction differences are -26.60, -24.08, -23.80, and -25.24 percentage points in streams 0–3, with all four paired intervals below zero. Table S20 reports every declared margin. These margins assess sensitivity; they are not validated application-specific utility thresholds.

The direct $H - U$ overall mean-fidelity differences at 50% retention are 0.01084, 0.00977, 0.00748, and 0.01015, with all four paired intervals above zero (Table S21). These signed-fidelity comparisons retain all gains and losses, irrespective of the harm margin. They show value within the specified batch-selection task, without implying the same gain for a calibration-only deployment policy.

6.4 Continuous estimation and the scope of template use

The original A radially projects an unconstrained weighted least-squares fit onto the Bloch ball. That projection need not minimize the constrained loss. R4 therefore adds its global constrained minimizer A^*_{fit} and a full-record control A^*_{all}, both solving

$$\min_{\|\boldsymbol{x}\|\le 1} \sum_{k\in\mathcal{I}} n_k \left(\boldsymbol{d}_k^{\mathsf{T}}\boldsymbol{x} - y_k\right)^2, \tag{35}$$

with $\mathcal{I}$ equal to the fitting labels or all eight labels. The latter uses the same total prepared-copy budget as the residual gate and accounts for the opportunity cost of holding measurements aside. Feasibility, optimality conditions, and loss improvement over A are checked for every fit. Optimization concerns this specified count loss, not the unknown target fidelity.

At 50% retention, the H ranking has higher grid-mean fidelity than A^*_{fit} in each stream, with differences 0.00522, 0.00815, 0.00676, and 0.00847 and paired intervals above zero. However, its differences from A^*_{all} are -0.01195, -0.00897, -0.01000, and -0.00949, with intervals below zero (Supplementary Table S19). Thus useful selection within a fitting-stage template task does not establish an advantage over continuous estimation using all available measurements.

A discrete template is an optional, restricted output in this study. A finite action alphabet could motivate such a decision in an application, but we have not measured a hardware cost or downstream utility that would justify that restriction. When an unrestricted continuous estimate and target fidelity are the only objectives, the full-record continuous control is the supported choice on this grid. The results characterize when residual evidence helps select template proposals; they do not recommend template substitution as a general recovery procedure.

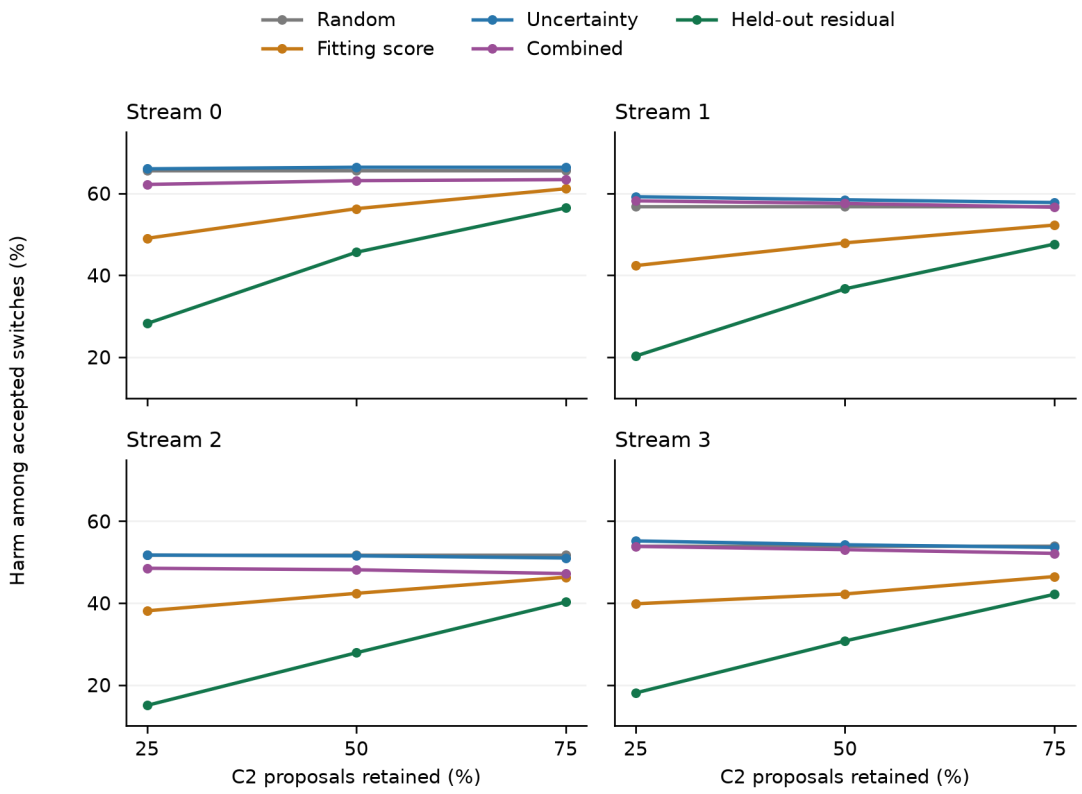


Figure 4. R4 selection quality at identical expected switching rates. Each point equally averages the harmful fraction among accepted switches over all 24 cells of one stream. Retention is relative to the original C_2 proposal set, so every rule has the same expected number of switches within a cell. Lines show point estimates; paired intervals and the distinction between original fixed gates and batch rankings are given in Supplementary Tables S17–S18. The plotted harm criterion is $F_B < F_A - 10^{-12}$; margin sensitivity and direct mean-fidelity comparisons are in Tables S20–S21. These are secondary analyses of the R3 records.

7 Conclusion

The audit found that the submitted clock construction was degenerate and that its reported veto effect could not be attributed to admissibility. We retain those results as an audit. The revision adds an energy-matched finite-clock construction and a template-switching experiment with separate calibration and held-out target sets.

The supported result concerns selection quality in a specified two-candidate task. After matching expected switching rates, held-out residual evidence selects fewer harmful template substitutions than random retention or uncertainty alone in the reported stream averages. This adds evidence beyond the unconditional harmful-switch reduction guaranteed by a one-sided veto.

The extra complexity of the combined gate is therefore not generically earned relative to uncertainty-only selection. Across streams 0–3, its unconditional harmful-switch-rate interval is strictly below that of the uncertainty-only gate in only 12/24, 8/24, 12/24, and 6/24 evaluated cells, respectively, and its mean-fidelity difference from the continuous estimate changes sign between streams. R4 finds a smaller but measurable conditional-quality benefit over U at matched coverage in the active 1,024-copy cells; nevertheless, residual-only batch ranking is better than the combined ranking in the stream averages. These comparisons do not justify preferring the extra uncertainty conjunct within the tested scaling family.

All-measurement continuous fitting gives higher grid-mean fidelity than the tested switching rankings, so the study provides no general reason to prefer a discrete template. The finite clock supplies a consistent physical realization of the count model, with no demonstrated necessity for the selection rule. The uncertainty statistic remains an uncalibrated proxy. R4 reuses the four R3 streams of pure logical-qubit targets and its intervals are pointwise and conditional on the fixed banks, known observation model, and thresholds. Mixed targets, model misspecification, out-of-grid generalization, an application-specific reason for discrete outputs, and channel-level recovery require separate evidence.

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Competing interests
The authors declare no competing interests.

Data availability
The R3 source code, protocol used for the runs, raw outputs, and submitted-grid audit artifacts are available at https://github.com/Jeong-Jaeyun/Risk-Aware-Relational-Spectral-Recovery-under-Partial-Quantum-Observability_Phase_1/tree/submission/revision-r3-clean. The R4 analysis protocol, source, input records, and complete cell summaries are supplied in the accompanying reproducibility archive.

Author contributions
J.Y.J. conceived the study, implemented the simulations, performed the data analysis, and drafted the manuscript. H.Y.J. supervised the research, provided critical intellectual input, and revised the manuscript. All authors have read and approved the final version of the manuscript.

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